```
GNU nano 8.7                                    opc.c                                    Modified
#include <stdio.h>

int main() {
    int pages[] = {1,2,3,4,1,2,5,1,2,3,4,5};
    int frames[3];
    int n = 12;
    int faults = 0, hits = 0;
    int i,j,k,pos,found,far;

    for(i=0;i<3;i++)
        frames[i] = -1;

    for(i=0;i<n;i++) {

        found = 0;

        for(j=0;j<3;j++) {
            if(frames[j] == pages[i]) {
                hits++;
                found = 1;
                break;
            }
        }

        if(!found) {

            pos = -1;
            far = i;

            for(j=0;j<3;j++) {

                int next = -1;

                for(k=i+1;k<n;k++) {
                    if(frames[j] == pages[k]) {
                        next = k;
                        break;
                    }
```

```
^G Help      ^O Write Out  ^F Where Is   ^K Cut      ^T Execute    ^C Location   M-U Undo     M-A Set Mark
^X Exit      ^R Read File  ^\ Replace    ^U Paste    ^J Justify    ^/ Go To Line M-E Redo     M-6 Copy
```

```
GNU nano 8.7                                    opc.c                                    Modified
                int next = -1;

                for(k=i+1;k<n;k++) {
                    if(frames[j] == pages[k]) {
                        next = k;
                        break;
                    }
                }

                if(next == -1) {
                    pos = j;
                    break;
                }

                if(next > far) {
                    far = next;
                    pos = j;
                }
            }

            if(pos == -1)
                pos = 0;

            frames[pos] = pages[i];
            faults++;
        }
    }

    printf("\nOptimal Page Replacement\n");
    printf("Page Faults = %d\n", faults);
    printf("Page Hits = %d\n", hits);

    printf("Hit Percentage = %.2f%%\n", (float)hits/n*100);
    printf("Fault Percentage = %.2f%%\n", (float)faults/n*100);

    return 0;
}
```

```
^G Help      ^O Write Out  ^F Where Is   ^K Cut      ^T Execute    ^C Location   M-U Undo     M-A Set Mark
^X Exit      ^R Read File  ^\ Replace    ^U Paste    ^J Justify    ^/ Go To Line M-E Redo     M-6 Copy
```

```
MAYUR LAP MSYS ~
$ nano opc.c

MAYUR LAP MSYS ~
$ gcc opc.c -o opc

MAYUR LAP MSYS ~
$ ./opc

Optimal Page Replacement
Page Faults = 7
Page Hits = 5
Hit Percentage = 41.67%
Fault Percentage = 58.33%

MAYUR LAP MSYS ~
$
```